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PARTS LIST

Semiconductors:

IC1	- NE555 timer
IC2	- CD4017 decade counter
IC3	- CD4001 quad NOR gate
IC4, IC5	- CD4081 quad AND gate
IC6, IC7	- CD4069 hex inverter
IC8	- 7812, 12V regulator
T1	- BC547 npn transistor
D1-D5	- 1N4148 switching diode
D6	- 1N4007 rectifier diode
BR1	- 1A bridge rectifier
LED1-LED13	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 15-kilo-ohm
R2	- 22-kilo-ohm
R3, R4	- 10-kilo-ohm
R5-R17	- 1-kilo-ohm
VR1	- 1-mega-ohm preset
VR2	- 10-kilo-ohm preset

Capacitors:

C1	- 100 μ F, 35V electrolytic
C2	- 0.01 μ F ceramic disk
C3	- 1000 μ F, 50V electrolytic
C4	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 15V, 500mA secondary transformer
S1	- On/off switch
S2	- Push-to-on switch

Again, pin 14 of CD4017 receives a clock pulse and its Q2 output goes high while the other outputs are low. The low Q0 and Q1 outputs are also connected to input pins 1 and 2 of NOR gate N1, respectively. Both the low inputs of NOR gate N1 make its output high and the red LED (LED2) glows. At the same time, both Q2 and Q3 outputs of CD4017 make the output of AND gate N9 high and the green LED glows (say, for road 2). All the three road lights are red. This light sequence for roads works one after the other as explained above.

When Q9 output of CD4017 goes high, counter CD4017 resets and the process restarts. The green and red lights glow for the same time period (say, one minute), which is set by preset VR1. The orange light glows for a few seconds (nearly 10 seconds).

When the orange light glows, transistor T1 receives base bias. The high at

the base of transistor T1 drives it into saturation and preset VR2 connects to pin 7 of NE555. Resistances of presets VR1 and VR2 come into parallel and the time period becomes small (nearly 10 seconds).

When there is a heavy rush on the current road, the traffic policeman can keep the traffic moving with green light by closing switch S1. NE555 doesn't regenerate clock pulse. As there is no clock signal for counter CD4017, its output does not advance. Open switch S1 for normal operation.

Fig. 2 shows the typical traffic light system on the road crossing.

Fig. 3 shows the power supply circuit. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 15V, 500 mA. The transformer output is rectified by a full-wave rectifier BR1, filtered by capacitor C3 and regulated by IC 7812 (IC8). Capacitor C4 bypasses any ripple present in the regulated 12V supply.

LED13 acts as the power-'on' indicator. Resistor R17 limits the current. A capacitor above 10 μ F is connected across the output of the regulator IC, while diode D6 protects the regulator IC in case its input is short to ground.

Construction

An actual-size, single-side PCB for the simple traffic light system is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a printed circuit board (PCB) to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Using preset VR1 set the 'glowing' time period for red and green lights as one minute. Using preset VR2 set the 'glowing' time period for the orange light as ten seconds. Connectors 'A' through 'L' provided on the PCB allow for extension of LEDs. For higher brightness, you can add more LEDs through relays. ●